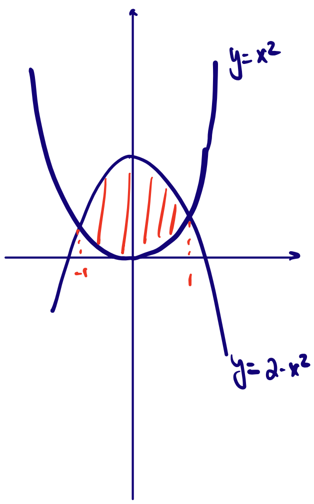
1. For each of the following improper integral, one may use Comparison Test to determine if they are convergent or divergent. Circle the correct answers to complete the arguments of using the comparison test.

1). The improper integral $\int_1^{\infty} \frac{1}{\sqrt[7]{x^5 - 0.9}}\, dx$ is **Convergent** or **Divergent** (circle the correct answer) because $\frac{1}{\sqrt[7]{x^5 - 0.9}}$ ___A___ ___E___ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^5}$, D). $\frac{1}{x^7}$, E). $\frac{1}{x^{5/7}}$, F). $\frac{1}{x^{7/5}}$

2. Sketch and shade the region bounded by the parabolas $y = x^2$ and $y = 2 - x^2$. Then find the area of the region.



$\begin{cases} y = x^2 \\ y = 2 - x^2 \end{cases} \Rightarrow x^2 = 2 - x^2 \Rightarrow 2x^2 = 2 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$

$$\text{Area} = \int_{-1}^{1} (2 - x^2) - x^2 \, dx$$

$$= \int_{-1}^{1} (2 - 2x^2)\, dx$$

$$= \left(2x - \frac{2}{3}x^3\right)\Big|_{-1}^{1}$$

$$= \left(2 - \frac{2}{3}\right) - \left(-2 + \frac{2}{3}\right) = \frac{8}{3}$$

Formula Sheet

$$\int_1^{\infty} \frac{1}{x^p}\, dx = \begin{cases} \text{is convergent} & \text{if } \quad p > 1 \\ \text{is divergent} & \text{if } \quad 0 < p \leq 1. \end{cases}$$